Good Church numerals as an ordered commutative rig

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Abstract

This directory gives a Rocq-checked relational interpretation of a commutative unital semiring with cancellative addition, a compatible strict partial order, and every standard finite-downset axiom. The construction uses Church numerals carrying fixed-rank induction certificates in the original pure type system PTS-rk. Equality is a state-indexed diagonal-observer relation, and order is a negative existential over certified differences. All source axioms and an independently defined classical first-order deduction system translate to inhabitants of the original typing judgment. The proof includes strict multiplication compatibility and a uniform construction for every cut-off, not merely successful finite examples. This note describes the definitions, proof terms, verification boundary, and the distinction between comparison with numerals and arbitrary linearity.

1 Statement and verification boundary

A *rig* is a semiring with a multiplicative unit; commutative refers to both operations. Here an ordered rig has a compatible **partial** order. Its strict part is preserved by addition and by multiplication with a strictly positive factor. Totality of the order is not part of the claim.

Theorem 1.1 (Formal interpretation)**.** *There is a relational interpretation in* PTS-rk *of a nontrivial commutative unital semiring with cancellative addition, zero least, monotone operations, strict addition and multiplication compatibility, and, for every external* $n \in \mathbb{N}$*,*

$$\forall x.\ x < \bar{n} \Longleftrightarrow \bigvee_{i<n} x = \bar{i}.$$

It also interprets all five schemes of relational Tarski–Mostowski–Robinson arithmetic R *and preserves an independent classical first-order deduction system. All these claims use one domain, one equality, one order, and the original falsity type.*

The public declaration in TheoremOrderedSemiring.v has the following unconditional Rocq type:

```
theorem_ordered_semiring :
  (forall A, Source.axiom A ->
    exists p,
      @typ nat RankSorts 0 ε p (sentence A)) /\
  (forall A, Source.derives [] A ->
    exists p,
      @typ nat RankSorts 0 ε p (sentence A)).
```

Here `sentence` is the structural translation of a closed source formula. `Source.axiom` and `Source.derives` are independently defined source predicates, not aliases for target inhabitation. The quantified proof term `p` belongs to the original annotated syntax `PTSrk.Syntax.Term nat 0`; `typ` and `RankSorts` are the original PTS rules.

The development neither adds PTS rules nor uses eta conversion, cumulativity, new primitives, axioms, admitted proofs, or annotation erasure. The reflective checker used for finite

proof-term certificates returns proofs of the original typing judgment. The final theorem does not replace that judgment by a checker acceptance relation. Numerical facts proved in Rocq select and justify actual PTS inhabitants; they do not stand in for those inhabitants.

Fresh compilation and recursive kernel checking on 20 October 2026 succeeded. The final `Print Assumptions` reports are closed under the global context. No external assertion that falsity is uninhabited is an assumption of these theorems. The precise internal proof/refutation statement is discussed in 8.1.

2 The unchanged calculus and arithmetic terms

The sorts are rk_i, for external natural numbers i, with $\mathsf{rk}_i : \mathsf{rk}_0$. Product formation uses the rank rule

$$\frac{\Gamma \vdash A : \mathsf{rk}_a \quad \Gamma, x : A \vdash B : \mathsf{rk}_b}{\Gamma \vdash (x : A) \to B : \mathsf{rk}_{\max(a+1,b)}}.$$

Conversion is beta conversion of the original annotated terms. In particular, a type of rank a is not automatically a type of rank $a + 1$.

Put $U := \mathsf{rk}_1$, so that $U : \mathsf{rk}_0$, and define

$$\mathsf{Void} := (Z : \mathsf{rk}_0) \to Z : \mathsf{rk}_1, \qquad \top := \mathsf{rk}_0 \to \mathsf{rk}_0 : \mathsf{rk}_1.$$

The closed term $\lambda(Z : \mathsf{rk}_0).Z$ inhabits $\top$. The implementation calls these types `Bottom` and `Top`.

The Church carrier is seed-first:

$$D := (X : U) \to X \to (X \to X) \to X : \mathsf{rk}_3.$$

With the lambda annotations determined by this type, the arithmetic operations are

$$\begin{aligned} 0 &:= \lambda X\, a\, f.a, \\ s\, n &:= \lambda X\, a\, f.f(n\, X\, a\, f), \\ \mathsf{Add}\, x\, y &:= \lambda X\, a\, f.x\, X\, (y\, X\, a\, f)\, f, \\ \mathsf{Mul}\, x\, y &:= \lambda X\, a\, f.x\, X\, a\, (\lambda u : X.y\, X\, u\, f). \end{aligned}$$

Write $1 := s\, 0$ and $\nu_n := s^n 0$. The source-oriented operations are deliberately reversed:

$$x \oplus y := \mathsf{Add}\, y\, x, \qquad x \otimes y := \mathsf{Mul}\, y\, x.$$

This convention matches the source arithmetic graphs and recursion equations. Commutativity is proved below; it is not silently used to change the definitions.

For proof types, use negative classical encodings:

$$\begin{aligned} \neg_c P &:= P \to \mathsf{Void}, \\ P \wedge_c Q &:= (P \to Q \to \mathsf{Void}) \to \mathsf{Void}, \\ \bigvee_{c,i<k} P_i &:= (P_0 \to \mathsf{Void}) \to \cdots \to (P_{k-1} \to \mathsf{Void}) \to \mathsf{Void}, \\ \exists_c x : A.\, P(x) &:= ((x : A) \to P(x) \to \mathsf{Void}) \to \mathsf{Void}. \end{aligned}$$

The empty disjunction is Void. A proof type P is *stable* when there is a term of type $\neg_c\neg_c P \to P$. The equalities, order predicates, and translated source formulas used below are stable. Negative existential witnesses are consumed inside continuations; the proof does not extract positive witness pairs.

3 Certified numerals and rank budgets

Definition 3.1 (Goodness certificate)**.** For the motive rank r, define

$$\mathsf{Good}_r(n) := (P : D \to \mathsf{rk}_r) \to P(0) \to ((u : D) \to P(u) \to P(s\,u)) \to P(n).$$

The certificate type has rank $r + 2$ for the budgets used here. Thus the subscript is the rank of the induction motive, not the rank of the certificate itself.

The zero and successor certificates are the ordinary Church induction constructors:

$$g_{r,0} := \lambda P\, p_0\, p_s.p_0, \qquad g_{r,s}\, n\, g := \lambda P\, p_0\, p_s.p_s\, n\, (g\, P\, p_0\, p_s).$$

RigCodes.v proves their native typing at the three fixed budgets 7, 11, and 15. These budgets are sufficient for the whole construction; no rank optimality is claimed.

Lowering a certificate requires an explicit lift of a lower-rank motive. Define

$$J_0 := \mathsf{rk}_0, \qquad J_{k+1} := J_k \to \mathsf{rk}_0, \qquad j_0 := \mathsf{rk}_0, \qquad j_{k+1} := \lambda(v : J_k).\mathsf{rk}_0.$$

Then $J_k : \mathsf{rk}_k$ and $j_k : J_k$. A motive value P of rank at most t can be lifted to exact rank t as $J_{t-1} \to P$. Induction into that lifted motive followed by evaluation at j_{t-1} gives the checked maps $\mathsf{Good}_{15} \to \mathsf{Good}_{11}$, $\mathsf{Good}_{11} \to \mathsf{Good}_7$, and $\mathsf{Good}_{15} \to \mathsf{Good}_7$. There is no converse promotion rule.

Lemma 3.2 (Closure)**.** *At each of the three budgets, the admitted objects are closed under zero, successor, explicit eta expansion, addition, and multiplication.*

Proof. Open the desired output certificate at an arbitrary predicate P and its zero/step proofs. For addition, apply the first input certificate to $u \mapsto P(\mathsf{Add}\,u\,y)$. Its zero case is P of the explicit eta expansion of y, obtained from the second input certificate. Its successor case uses the supplied step of P.

For multiplication, use $u \mapsto P(\mathsf{Mul}\,u\,y)$. At successor, the equation $\mathsf{Mul}(s\,u)y =_\beta \mathsf{Add}\,y\,(\mathsf{Mul}\,u\,y)$ lets a second induction on y supply the needed iteration of the step. The implementation in DomainClosure.v never inducts into Good_r itself and never transports a positive goodness certificate through the observational equality defined next. □

4 A single observer equality

For $A, B : U$, put

$$T(A, B) := (\neg_c A \to \neg_c B) \wedge_c (\neg_c B \to \neg_c A) : \mathsf{rk}_5.$$

An observer consists of data

$$X : U, \quad r : X \to X \to \mathsf{rk}_5, \quad a : X, \quad k : U \to X, \quad f : X \to X, \quad d : X \to (U \to X) \to$$

$$X, \quad p : X \to U,$$

together with the following nine certificates. All displayed object variables are universally bound at their indicated types; write $d_w h := d\,w\,h$.

$$
\begin{aligned}
&r(v,v),\\
&r(v,w) \to r(w,v),\\
&r(u,v) \to r(v,w) \to r(u,w),\\
&r(v,w) \to r(fv,fw),\\
&((z:U) \to r(hz,h'z)) \to r(d_w h, d_w h'),\\
&r(f(kz),kz),\\
&r(d_w k, w),\\
&r(d_w(\lambda z:U.v),v),\\
&r(v,w) \to T(pv,pw).
\end{aligned}
$$

Here $u,v,w : X$, $z : U$, and $h,h' : U \to X$. These certificates are **bound fields of each observer**, not global axioms.

Definition 4.1 (Diagonal-observer equality)**.** Let $\mathcal{A}$ denote the entire dependent telescope of observer data and certificates. Define

$$\mathsf{E}(x,y) := \mathcal{A} \to T\big(p(x\,X\,a\,f), p(y\,X\,a\,f)\big) : \mathsf{rk}_7.$$

The notation $\mathcal{A} \to B$ expands dependent arrows; it does not add a record or quotient type.

ObservationEquality.v proves stability, reflexivity, symmetry, transitivity, successor congruence, and observational eta equality. For example, successor congruence reuses the observer with p replaced by $p \circ f$. Eta expansion is observationally equal because its applications to X, a, f beta-reduce to the same expressions; eta conversion is not used.

ArithmeticCongruence.v proves the guarded addition and multiplication congruences. They are specific to these operations and their certified arguments. The development does not assume that every raw function $D \to D$ respects E.

4.1 Successor separation

Lemma 4.2 (Successor facts)**.** *There are closed terms realizing*

$$\mathsf{Good}_7(x) \to \mathsf{E}(s\,x,0) \to \mathsf{Void}, \qquad \mathsf{Good}_7(x) \to \mathsf{Good}_7(y) \to \mathsf{E}(s\,x,s\,y) \to \mathsf{E}(x,y).$$

Proof. SuccessorSeparation.v constructs two actual observers and supplies all nine certificates for each.

For nonzeroness, take $X := U \to U \to U$ and

$$
\begin{aligned}
r(v,w) &:= (A,B:U) \to T(v\,A\,B, w\,A\,B),\\
a\,A\,B &:= A, \qquad\qquad k\,C\,A\,B := C,\\
f\,v\,A\,B &:= v\,B\,B,\\
d\,w\,h\,A\,B &:= h(w\,A\,B)\,A\,B,\\
p\,v &:= v\,\top\,\mathsf{Void}.
\end{aligned}
$$

Its zero observation is $\top$. Induction using $\mathsf{Good}_7(x)$ identifies the successor observation with Void in the sense of T. An assumed $\mathsf{E}(s\,x,0)$ then gives a term of Void by applying the two negative implications to the closed inhabitant of $\top$.

For injectivity, start with an arbitrary observer and form a delayed one. Its carrier is $X' := U \to X$, with pointwise relation $r'(v, w) := (z : U) \to r(vz, wz)$, and data

$$\begin{aligned} a' &:= k, & k'\, z &:= \lambda u : U.k\, z,\\ f'\, v &:= \lambda u : U.d\, a\, (\lambda z : U.f(vz)),\\ d'\, w\, h &:= \lambda u : U.d(wu)(\lambda z : U.h\, z\, u),\\ p'\, v &:= p(v\, \top). \end{aligned}$$

The original nine certificates prove the nine delayed certificates. In particular, the state-indexed argument of d is retained. The type $U \to X$ still has rank one because $U : \mathsf{rk}_0$.

Write $F_n := n\, X\, a\, f$ and $F'_n := n\, X'\, a'\, f'$. A rank-five simulation motive, lifted to rank seven, proves

$$\mathsf{Good}_7(n) \to (z : U) \to r(F'_{s\,n} z, F_n).$$

At zero use $f(kz) \sim kz$ and $d_a k \sim a$; at successor use pointwise congruence of d_a and its law on constant functions. Apply the assumed successor equality to the delayed observer, then compose the two simulations via the p bridge. This recovers the result of every original observer and hence $\mathsf{E}(x, y)$. □

This equality is not the rank-five predicate equality of the older relational R library. Only the common carrier, arithmetic beta computations, syntax, and proof infrastructure are reused. Facts about the old equality or order are not renamed into facts about the new E and O.

5 Retained induction and the rig laws

Lemma 5.1 (Induction retaining a lower certificate)**.** *Let $P(u)$ be stable, of rank seven, eight, or nine. Given closed realizers of*

$$P(0), \qquad (u : D) \to \mathsf{Good}_7(u) \to P(u) \to P(s\, u),$$

one obtains a realizer of $(x : D) \to \mathsf{Good}_{11}(x) \to P(x)$.

Proof. Use the rank-eleven invariant $I(u) := \mathsf{Good}_7(u) \wedge_c P(u)$. The base pairs the zero certificate with the base proof. To make the successor pair, open the previous negative pair inside a continuation, apply the goodness successor constructor and the supplied step, and feed both outputs to the new continuation. Induction using $\mathsf{Good}_{11}(x)$ gives $I(x)$. Project $P(x)$ negatively and use its stability. The proof never extracts a positive Good_7 certificate outside its continuation. The three native instances are checked in RetainedInduction.v. □

It is useful to distinguish the larger domain $W = (D, \mathsf{Good}_{11})/\mathsf{E}$ from the main domain $V = (D, \mathsf{Good}_{15})/\mathsf{E}$. These are descriptions of a guarded first-order interpretation, not internal quotient constructions.

Associativity of Add and Mul, and

$$\mathsf{Mul}(\mathsf{Add}\, x\, y)z =_\beta \mathsf{Add}(\mathsf{Mul}\, x\, z)(\mathsf{Mul}\, y\, z),$$

are raw beta identities. Some unit equations return an explicit eta expansion, handled by observational eta equality.

Rank-seven induction first proves the successor-shift law for addition and then addition commutativity. It also supplies the remaining multiplication-zero and unit laws. Retained induction proves

$$\mathsf{Good}_{11}(x) \to \mathsf{Good}_7(y) \to \mathsf{E}(\mathsf{Mul}\, x\, (s\, y), \mathsf{Add}\, x\, (\mathsf{Mul}\, x\, y)).$$

The successor step needs addition congruence, associativity and commutativity; the retained Good_7 certificate justifies these uses. A second retained induction proves multiplication commutativity on W. The other distributive law follows. SemiringLaws.v supplies the closed inhabitants of these generic equations.

Lemma 5.2 (Cancellation, zero sums and zero products)**.** *On W, addition is cancellative, and*

$$\mathsf{E}(x \oplus y, 0) \to \mathsf{E}(x, 0) \wedge_c \mathsf{E}(y, 0), \qquad \mathsf{E}(x \otimes y, 0) \to \mathsf{E}(x, 0) \vee_c \mathsf{E}(y, 0).$$

Proof. For cancellation, use 5.1 with the rank-eight motive

$$P(u) := \mathsf{E}(\mathsf{Add}\, u\, x, \mathsf{Add}\, u\, y) \to \mathsf{E}(x, y).$$

The successor step applies successor injectivity to the two sums and then the induction hypothesis. The sums have the lower goodness certificates needed for injectivity.

For zero sums use $P(u) := \mathsf{E}(\mathsf{Add}\, u\, y, 0) \to \mathsf{E}(u, 0)$. The zero case is reflexivity; the successor case contradicts successor nonzeroness. Commutativity supplies the other summand.

For zero products fix y and use the rank-nine stable motive

$$P(u) := \mathsf{E}(\mathsf{Mul}\, u\, y, 0) \to (\mathsf{E}(u, 0) \vee_c \mathsf{E}(y, 0)).$$

At zero choose the first disjunct. At successor, the equation for $\mathsf{Mul}(s\, u)y$ and zero-sum reflection give $\mathsf{E}(y, 0)$. This step needs the retained certificate for u but not the induction hypothesis itself. All three constructions are checked in Cancellation.v. □

6 Natural difference order and strictness

Definition 6.1 (Order)**.** For arbitrary $x, y : D$, let

$$\mathsf{O}(x, y) := ((c : D) \to \mathsf{Good}_{11}(c) \to \mathsf{E}(\mathsf{Add}\, c\, x, y) \to \mathsf{Void}) \to \mathsf{Void} : \mathsf{rk}_{15}.$$

Write $x \leq y$ for $\mathsf{O}(x, y)$ and define $x < y := \mathsf{O}(x, y) \wedge_c \neg_c \mathsf{E}(x, y)$. The strict predicate has rank seventeen.

The order asserts a negative existential difference c satisfying $x \oplus c =_E y$. Crucially, this witness has a Good_{11} certificate even when the endpoints belong to V. The order on V is inherited from W.

Lemma 6.2 (Compatible strict partial order)**.** *On W, the relation $\leq$ is reflexive, transitive and antisymmetric modulo E. Zero is least, both operations are monotone, and their strict laws include*

$$x < y \to x \oplus z < y \oplus z, \qquad x < y \to 0 < z \to x \otimes z < y \otimes z.$$

The corresponding left-argument laws hold too. All these laws restrict to V.

Proof. Reflexivity chooses difference zero; zero leastness chooses the object itself. Transitivity adds the two differences, using closure. If $x \oplus c =_E y$ and $y \oplus d =_E x$, cancellation gives $c \oplus d =_E 0$, and zero-sum reflection gives $c =_E 0$. Hence $x =_E y$.

Additive monotonicity keeps the same difference. Multiplicative monotonicity uses $c \otimes z$ and distributivity. Closure supplies the witness certificate in each case. Strict additive monotonicity follows from cancellation.

For strict multiplication, suppose the products were equal. Distributivity and cancellation would give $c \otimes z =_E 0$. The zero-product disjunction gives $c =_E 0$ or $z =_E 0$. The first contradicts $x < y$ using the difference equation; the second contradicts $0 < z$. This additional argument is needed: ordinary monotonicity alone does not imply strict multiplication compatibility.

Every witness is consumed within the negative existential continuation. The final conclusions are stable. The definitions and native proof terms are in NaturalOrder.v and OrderLaws.v. □

7 Every finite cut at one fixed rank

For an external cutoff n, put

$$K_n(x) := (t : D) \to \mathsf{Good}_{11}(t) \to \mathsf{E}(x, s^{n+1}t) \to \mathsf{Void},$$
$$\mathsf{Cut}_n(x) := (\mathsf{E}(x, \nu_0) \to \mathsf{Void}) \to \cdots \to (\mathsf{E}(x, \nu_n) \to \mathsf{Void}) \to K_n(x) \to \mathsf{Void}.$$

The rank of $K_n(x)$ is fourteen and that of $\mathsf{Cut}_n(x)$ is fifteen, independently of n. The large case is flattened into the branch telescope. Nesting ordinary binary disjunctions instead would waste rank as the cutoff grows.

Lemma 7.1 (Uniform finite cut)**.** *For every external natural number n, there is a closed original PTS inhabitant of*

$$(x : D) \to \mathsf{Good}_{15}(x) \to \mathsf{Cut}_n(x).$$

Proof. Induct directly into $u \mapsto \mathsf{Cut}_n(u)$ with the supplied Good_{15} certificate. After opening the branch continuations, the zero case selects the branch $u =_E \nu_0$. For successor, a finite branch $u =_E \nu_i$ with $i < n$ goes to the next finite branch. The last finite branch goes to the large branch with witness zero. A large branch $u =_E s^{n+1}t$ goes to the large branch with witness $s\,t$ and its successor certificate. Successor congruence supplies all equality steps. The case $n = 0$ has no first group of finite successor branches.

FiniteCuts.v constructs the telescope and its inhabitant recursively in the external parameter n. Its theorem `finite_cut_realized` is quantified over every n; it is not a finite test suite. □

A finite branch $x =_E \nu_i$, $i \le n$, gives $x \le \nu_n$ with difference ν_{n-i}. In a large branch, $x =_E s^{n+1}t$ gives $\nu_n \le x$ with difference $s\,t$. If it also satisfied $x \le \nu_n$, combining the difference equations and cancelling n successors would give a successor equal to zero. The intermediate sums carry the necessary Good_7 certificates by closure and lowering.

Consequently, for every n, the main domain realizes

$$x \le \nu_n \iff \bigvee_{c,i \le n} \mathsf{E}(x, \nu_i), \qquad x \le \nu_n \vee_c \nu_n \le x.$$

These are BoundedCases.v's all-index results. Numeral separation removes the endpoint from the first equivalence to give

$$x < \nu_n \iff \bigvee_{c,i<n} \mathsf{E}(x, \nu_i).$$

StrictBounded.v also proves the bridge to the *literal* source sentence, including the empty downset at $n = 0$. The bridge uses beta conversion where the source expands strict order but the target applies the closed strict-order constructor.

For reference, the main fixed ranks are

type	D	$\mathsf{E}(x,y)$	$\mathsf{Good}_7(x)$	$\mathsf{Good}_{11}(x)$	$\mathsf{Good}_{15}(x)$	$\mathsf{O}(x,y)$	$x < y$	$\mathsf{Cut}_n(x)$
rank	3	7	9	13	17	15	17	15

8 The source theory and preservation of deductions

Source.v uses an independent first-order syntax whose term symbols are zero and successor. Addition and multiplication are ternary relation symbols $A(x, y, z)$ and $M(x, y, z)$, not target-language functions smuggled into source terms. Equality, $\leq$, falsity, implication and universal quantification are primitive formula constructors. Other connectives are the negative abbreviations already displayed.

The finite list `Source.rig_laws` contains exactly 23 universally closed relational axioms:

1. For each of A and M: totality, uniqueness, commutativity and associativity (eight axioms).
2. Additive zero, multiplicative zero and unit, distributivity, and additive cancellation (five axioms).
3. Order reflexivity, transitivity, antisymmetry, zero leastness and $0 \neq 1$ (five axioms).
4. Monotonicity of A and M, strict additive monotonicity and strict multiplicative monotonicity with a positive factor (four axioms).
5. Successor as addition of one: $A(x, 1, s\,x)$ (one axiom).

For example, relational associativity of A is the universal closure of

$$A(x, y, u) \to A(u, z, w) \to A(y, z, v) \to A(x, v, w).$$

Totality and uniqueness make this the familiar associative operation law without changing the relational source signature. Commutativity supplies the omitted opposite-side unit and monotonicity equations.

The other source axiom constructors contain the strict finite-cut schema and the five relational R schemes, for all external m, n:

$$\begin{aligned}
&\forall z.\ A(\bar{m}, \bar{n}, z) \iff \overline{m + n} = z,\\
&\forall z.\ M(\bar{m}, \bar{n}, z) \iff \overline{mn} = z,\\
&\bar{m} \neq \bar{n} \quad (m \neq n),\\
&\forall x.\ x \leq \bar{n} \iff \bigvee_{i \leq n} x = \bar{i},\\
&\forall x.\ x \leq \bar{n} \vee \bar{n} \leq x.
\end{aligned}$$

All formula connectives here denote the declared source abbreviations. The source theory contains no target-typing premise and no assumption of its own interpretation.

The structural translation sends the atomic formulas to

$$\begin{aligned}(x = y)^* &:= \mathsf{E}(x^*, y^*),\\ (x \le y)^* &:= \mathsf{O}(x^*, y^*),\\ A(x, y, z)^* &:= \mathsf{E}(x^* \oplus y^*, z^*),\\ M(x, y, z)^* &:= \mathsf{E}(x^* \otimes y^*, z^*).\end{aligned}$$

Falsity goes to Void, implication to the ordinary arrow, and universal quantification is guarded:

$$(\forall x.\, B)^* := (x : D) \to \mathsf{Good}_{15}(x) \to B^*.$$

FirstOrderTranslation.v proves typing, stability, and compatibility with renaming and substitution. Its formula rank is computed recursively: equality and arithmetic atoms have rank seven, order atoms rank fifteen, falsity rank one, implication has rank $\max(a + 1, b)$, and universal quantification has rank $\max(18, b)$.

The source deduction rules are assumptions, theory axioms, implication introduction and elimination, universal introduction and elimination, double-negation elimination, equality reflexivity and equality replacement. FirstOrderEquality.v proves the replacement operation by structural induction on formulas, using the guarded arithmetic congruences. It does not identify observer equality with definitional equality.

Proof of Theorem 1.1. RigAxiomRealization.v realizes the actual translations of all 23 relational rig axioms, including their Good_{15} quantifier guards. RAxiomRealization.v realizes every instance of the five R schemes. StrictBounded.v supplies every strict finite-cut sentence.

Induction on the independent source derivation now constructs the target inhabitant. Implication uses native abstraction/application; quantification uses the guarded translation and its substitution theorem; the classical rule uses proved stability. Equality replacement uses its separately proved structural theorem. The general lemma in FirstOrderDeduction.v takes an axiom-realization parameter. The public theorem supplies the actual realizers above, so no such parameter remains in its type. □

The public all-cutoff and comparison declarations make the quantification explicit:

```
bounded_cases cutoff :
  exists p, @typ nat RankSorts 0 ε p
    (sentence (Source.strict_bounded cutoff)).

numeral_comparison cutoff :
  exists p, @typ nat RankSorts 0 ε p
    (sentence (comparison_sentence cutoff)).
```

8.1 Ground truth and internal refutation

GroundFaithfulness.v proves the correct ground arithmetic beta computations and constructs proof or refutation inhabitants according to each numerical test. For example, $m + n = k$ gives a proof of $\mathsf{E}(\nu_m \oplus \nu_n, \nu_k)$, whereas $m + n \neq k$ gives a proof of its negation. Likewise $i \le j$ gives $\mathsf{O}(\nu_i, \nu_j)$, and $j < i$ gives its negation. The results cover all

closed source terms and all equality, order, addition and multiplication atoms, not just hand-selected numerals.

These are **internal refutations**: a false atom has a term mapping any claimed proof of that atom to Void. They do not, by themselves, assert externally that Void has no inhabitants. Turning internal refutation into an external non-inhabitation statement would require that distinct metatheoretic result. Neither the interpretation nor its preservation proof assumes such a result or uses a set model.

9 What is not claimed: predecessors and linearity

At cutoff zero, the finite-cut theorem gives zero or a successor of an object carrying Good_{11}. It does **not** provide a predecessor with Good_{15}, and it supplies no promotion from the weaker to the stronger certificate. Thus the order on V is not claimed to be the natural difference order formed using differences inside V itself.

Graphs.v additionally checks native translations of Robinson's Q1, Q2, Q4, Q5, Q6 and Q7 using the same equality and main guard. It does not prove Q3, the same-domain predecessor axiom, or an interpretation of full Robinson arithmetic Q. Those six equations are not silently inserted as additional constructors of the public `Source.axiom` predicate.

Comparison with each fixed standard numeral is a genuine weak comparison principle. Arbitrary totality, trichotomy, or strict cotransitivity remain unproved for this particular PTS interpretation. They also do not follow merely from the stated source algebra and order laws.

Remark (A nonlinear source model). Let C consist of zero and those polynomials in $\mathbb{Z}[t]$ that are eventually positive as t tends both to $+\infty$ and to $-\infty$. Equivalently, its nonzero elements are positive integer constants and positive-leading-coefficient polynomials of positive even degree. Use the ordinary polynomial operations and define

$$p \le q \quad \Longleftrightarrow \quad q - p \in C.$$

Then C is a cancellative commutative unital semiring with no zero divisors. This is its natural difference order: zero is least, addition preserves strict inequality, and multiplication by a nonzero element does too.

Every nonconstant member exceeds every standard integer. The elements below a fixed numeral n are exactly $0, \ldots, n$, so all the finite downset and numeral-comparison schemes hold. Successor is $p \mapsto p + 1$. This model even has same-domain predecessors: subtracting one from a nonconstant member leaves it in C.

Nevertheless $t^2 + t$ and $t^2 - t$ are incomparable, since their nonzero differences have odd degree. Thus totality, trichotomy, and the principle that mutual absence of strict inequality implies equality all fail. Strict cotransitivity fails as well: for $x = t^2$, $y = t^2+t$, and $z = t^2+1$, one has $x < z$ but neither $x < y$ nor $y < z$.

This is a mathematical source-theory counterexample, not a Rocq-verified set model and not a model of every PTS type. It does not show that stronger comparison is impossible in the specific Church/observer interpretation. Any such improvement would need additional properties of that interpretation, beyond the source laws proved here.

10 Files and reproduction

The public library name is `OrderedSemiring`. Its proof layers are arranged as follows:

- RigCodes.v, DomainClosure.v: the three goodness budgets, lowering, and native closure.
- ObservationEquality.v, ArithmeticCongruence.v, SuccessorSeparation.v: the new observer equality and successor facts.
- RetainedInduction.v, SemiringLaws.v, Cancellation.v: generic algebra, cancellation and zero-product cases.
- NaturalOrder.v, OrderLaws.v: the difference order and strict compatibility.
- FiniteCuts.v, OrderNumerals.v, OrderCuts.v, BoundedCases.v, StrictBounded.v: uniform finite cuts and their source-level bridges.
- ModelInterface.v, Source.v, FirstOrderTranslation.v, FirstOrderEquality.v: the independent source and structural translation.
- RigAxiomRealization.v, RAxiomRealization.v, FirstOrderDeduction.v, TheoremOrderedSemiring.v: all-index axiom realization and deduction preservation.
- GroundFaithfulness.v, Graphs.v: ground proofs/refutations, total functional graphs and the six additional native Robinson translations.

Dependencies are the unchanged `PTSrk` kernel, the common `Interpretation_of_R` infrastructure, and its independent `Interpretation_of_R.relational` library. There is no dependency on the functional interpretation, the set-theoretic model, or a new normalization assumption. Only the required common and relational infrastructure accompanies this development.

From the repository root, run

```
bash theories/OrderedSemiring/check.sh
```

The checker records tool versions and exact logs, builds in a fresh absolute Dune directory so that assumption reports are regenerated, then recursively kernel-checks the final theorem and its dependencies. The verified environment used Rocq 9.2.0, Stdlib 9.1.0, Equations 1.3.2+9.2, OCaml 4.14.2 and Dune 3.23.1. Consult check.sh for toolchain selection and log-directory overrides; no package installation is performed by the check.

This note is itself LaTeX–Markdown source, using the same document dialect as the main paper. To regenerate its PDF, run

```
bash theories/OrderedSemiring/doc/build.sh
```

The bundled preamble.tex and document runtime make this a local documentation build. Neither a hosted prose service nor an external bibliography or proof badge is needed. The result is README.pdf.

The mathematical starting point was the supplied note *Good Church numerals as an ordered commutative rig* of 14 October 2026. The Rocq development goes beyond its experimental checks: it supplies the uniform all-cutoff proof, strict multiplication compatibility, independent deduction preservation, and fresh kernel evidence. Its source revision is 5530d1e.